

Paper 7.50 - Discrete-Continuous Bridge Claim Contract

CQE-paper-07.50

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-07.50.md

Paper 7.50 - Discrete-Continuous Bridge Claim Contract

Purpose

Paper 7.50 defines what counts as a valid bridge claim. It preserves the discrete proof while allowing continuous presentation.

Admitted Paper 7 Claims

The following claims are admitted from Paper 7:

```
finite numeric traces admit piecewise-linear interpolation
integer samples are preserved exactly
maximum sample error is zero
adjacent segments share endpoints
Rule30/Rule90 correction identity remains true on indexed LCR states
between-sample physics remains an obligation
```

Claim Requirements

Any later paper using a Paper 7 bridge must state:

```
the original indexed trace
the interpolation rule
the sample preservation receipt
whether any between-sample claim is being made
which separate receipt proves any between-sample claim
```

Linked Receipt

The minimum receipt link for Paper 7 is:

```
paper: CQE-paper-07
theorem: Sample-preserving discrete-continuous bridge
receipt: production/formal-papers/CQE-paper-07/discrete_continuous_bridge_receipt.json
status: pass
```

The receipt is sufficient for sample preservation. It is not sufficient by itself for Hamiltonian dynamics, continuum collapse, CMB interpretation, Hawking claims, or any physical uniqueness between samples.

Boundary Failures

The following are boundary failures:

```
erasing the original discrete receipt after drawing the bridge
claiming a curve proves physical dynamics between samples
changing indexed sample values without a new receipt
using interpolation to hide correction or residual rows
importing continuous-field language without naming the trace
```

Boundary failures become obligations or are rejected by the consuming paper.

Hidden-Guess Contract

When hidden-guess diagnostics are enabled, the answer must be recorded before the receipt is

revealed. The guess record should include:

```
prompt  
predicted sample error or scope status  
revealed receipt  
match/mismatch  
next obligation if mismatch
```

The between-sample overclaim prompt is required because the correct answer is negative.

Conclusion

Paper 7.50 lets later papers use continuous presentations honestly. It admits sample preservation while keeping physical dynamics tied to separate proof.